



Mercury has multiple, superposed global tectonic patterns

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ABSTRACT

There is a long tradition in studying tectonic landforms on Mercury, as it hosts myriad structures accommodating both shortening and extension. These landforms have been extensively studied in terms of size, distribution, and age, and many tectonic maps have been produced. The orientations of fault-related landforms were previously studied for subsets of structures, but a detailed global investigation of orientations for all tectonic landforms on Mercury has yet to be carried out. Variations in orientations of tectonic structures across the globe can provide insights into the tectonic processes that formed them. We compile a global tectonic map that contains 18,541 structures specified as shortening landforms, extensional landforms, or troughs. We process the map data to extract fault orientations in three different geographic configurations with radial and concentric bins around the rotational poles, the Caloris basin, and the hot poles. In each of these three representations, we find systematic tectonic patterns that, taken together, show multiple, superposed preferred orientations of faults. In particular, systematic thrust fault orientations radial and concentric to the Caloris basin are detected on the hemispheric scale, revealing an enormous impact damage zone where fractures were reactivated by later tectonic processes. Furthermore, faults display systematic orientations around the hot poles, showing that tectonic processes or lithospheric strength and thickness variations arising from Mercury's spin-orbit resonance with the Sun have played a role in its tectonic history.

1. Introduction

1.1. Tectonic landforms on Mercury

Mercury's geological evolution includes a rich history of tectonism, as evinced by thousands of tectonic landforms found across its surface (see overview in Byrne et al., 2018). Among the most prominent tectonic landforms are positive-relief shortening landforms and negative-relief landforms accommodating extension. Shortening landforms show a large geomorphological variability (Loveless et al., 2024) but generally comprise symmetric or asymmetric ridges that are often accompanied by scarps (Byrne et al., 2018, 2014; Loveless et al., 2024; Strom et al., 1975; Watters et al., 2001; Watters and Nimmo, 2009). The scarps represent surface breaks of thrust faults, with ridges being manifestations of the hanging-wall uplift that occurred along these thrusts (e.g., Byrne et al., 2018, 2014; Watters and Nimmo, 2009). Many of these landforms are hundreds of kilometers long and show a few thousands of meters of structural relief (e.g. Byrne et al., 2014; Watters et al., 2001). These landforms are found globally across Mercury (Byrne et al., 2014), and

their formation and growth are widely accepted to have been driven by global contraction, a process resulting from the long, sustained cooling of the innermost planet's interior (Hauck et al., 2004; Solomon, 1976; Tosi et al., 2013).

Landforms accommodating extension can be described as narrow troughs. The largest are thought to be grabens, as they are often accompanied by relay ramps, a structural geometry diagnostic of normal faults. Prominent grabens on Mercury are only found in impact basins that were volcanically flooded, such as the Caloris or Rembrandt basins (Strom et al., 1975; Watters et al., 2009, 2005). The largest of these grabens measure up to several tens of kilometers long and up to 8 km wide (Klimczak et al., 2013, 2010). The cooling and resulting horizontal contraction of the volcanic units within the basins is thought to have led to extension that was accommodated by normal faults and thus grabens (Blair et al., 2013; Freed et al., 2012; Klimczak et al., 2012). Small grabens can be found globally across Mercury (Man et al., 2023), where they form on or alongside the ridge crests of larger shortening landforms as part of hanging-wall deformation (Byrne et al., 2018; Klimczak et al., 2019; Man et al., 2023).

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There is a long tradition of tectonic mapping on Mercury. The return of the first photogeological data of Mercury from the Mariner 10 mission led to early maps of scarps, ridges, and lineaments that allowed for the characterization of the tectonic history of the planet (Dzurisin, 1978; Strom et al., 1975; Thomas et al., 1988). With improvements in Mariner 10 image processing and availability of the first regional stereotopographic data for Mercury, tectonic maps, especially of large-scale shortening landforms, were refined and analyzed for structure orientations, densities, and the strains they accommodate (e.g., Watters et al., 2004, 2001). The MErcury Surface, Space ENvironment, GEOchemistry, and Ranging (MESSENGER) mission returned four years of orbital observations that enabled the production of several global image mosaics (Denevi et al., 2017), as well as detailed topographic datasets (Bertone et al., 2023; Manheim et al., 2022; Fassett, 2016; Preusker et al., 2017; Zuber et al., 2012) that have facilitated tectonic mapping on local, regional, and global scales. These efforts include global maps of shortening landforms (Byrne et al., 2014; Man et al., 2023; Watters et al., 2015), complete tectonic maps of the Caloris (Byrne et al., 2018; Cunje and Ghent, 2016) and Rembrandt basins (Byrne et al., 2018; Ferrari et al., 2015), regional maps of shortening landforms in Borealis Planitia (Cardinale et al., 2023; Crane and Klimczak, 2019), maps of grabens situated within peak-ring basins (Byrne et al., 2018) and flooded craters (Klimczak et al., 2012), and a global map of straight-rim segments of polygonal impact craters (Yazıcı et al., 2024).

1.2. Tectonic processes and their predicted patterns on Mercury

Among the tectonic processes that have been studied and assessed for their predicted structural patterns on Mercury are global contraction, tidal despinning, planetary reorientation from the Caloris impact, and mantle convection. Global contraction is arguably the dominant tectonic process on Mercury. The long, sustained cooling of the planet that caused a reduction in planetary volume (e.g., Hauck et al., 2004; Solomon, 1976; Tosi et al., 2013) produces a global, contractional tectonic regime conducive to forming thrust faults. Global contraction operating in isolation will produce a horizontally isotropic stress state in the lithosphere that promotes random orientations of shortening landforms across Mercury's surface (Melosh and McKinnon, 1988). Several studies confirm that mapped orientations of shortening landforms on Mercury, even accounting for biases arising from fixed illumination geometries, are not random (Byrne et al., 2014; Crane and Klimczak, 2019; Watters et al., 2015), indicating that other processes must have played some part in explaining the observed orientations of tectonic structures there.

Mercury is in a 3:2 spin-orbit resonance with the Sun, with a sidereal rotational period of approximately 59 Earth days. It is likely that Mercury's rotation has been substantially tidally slowed (Burns, 1976). Such pronounced tidal despinning changes the planetary figure by reductions in both polar flattening and equatorial bulging that, in turn, induce a change in global stresses (Beuthe, 2010; Burns, 1976; Melosh, 1980; Melosh and Dzurisin, 1978; Melosh and McKinnon, 1988). Patterns arising from stresses induced by tidal despinning operating alone were predicted to cause latitudinal tectonic provinces with east-west-oriented normal faulting in the polar regions, mid-latitudinal strike-slip faulting, and north-south-aligned thrust faulting at the equator (Melosh, 1980; Melosh and McKinnon, 1988). For any faulting to occur, these predictions rely on the assumption that stress states are fully compressive. Given that the solutions contain tensile stress components, it must be considered that the overburden would quickly shift stresses to a fully compressive stress state with depth. Rock-mechanical reassessments of these stress solutions showed that a global pattern of jointing, a distinctly different fracturing type accommodating purely opening instead of shearing displacements, is more likely to have resulted from tidal despinning alone, with preferred east-west jointing predicted at the equator and randomly oriented joints at the poles (Klimczak et al., 2015).

If tidal despinning occurred late enough in Mercury's geological

history, it may have overlapped with global contraction, a scenario that has been previously discussed (Beuthe, 2010; Dombard and Hauck, 2008; Klimczak et al., 2015; Melosh and Dzurisin, 1978; Pechmann and Melosh, 1979). Predictions of tectonic patterns resulting from such overlap generally agree on north-south-oriented thrust faulting in the equatorial and mid-latitudinal regions of Mercury (Beuthe, 2010; Klimczak et al., 2015; Pechmann and Melosh, 1979), with north-south orientations becoming less pronounced poleward (Klimczak et al., 2015). More complicated scenarios involving tidal despinning overlapping with global contraction, but which also account for tidal deformation and reorientation caused by the Caloris impact, find a pattern of thrust faults oriented radially around the assumed locations of Mercury's rotational poles prior to that reorientation (Matsuyama and Nimmo, 2009).

Numerical simulations of convection within Mercury's mantle produce solutions that show long, linear up- and downwellings (King, 2008) that were interpreted to be consistent with the orientations and ages of shortening structures known at that time (Watters et al., 2004). Later convection models incorporating improved knowledge of the internal structure of Mercury from MESSENGER observations found smaller convection cell geometries (Tosi et al., 2013) that are unlikely to produce systematic patterns on a global scale. Nevertheless, correlations of thick crust with the areal distribution and strain of a subset of Mercury's shortening landforms were taken to indicate that mantle convection may play a role in the localization and orientation of these structures after all (Watters et al., 2021, 2015).

1.3. Motivation of this study

The orientations of faults have previously been studied for shortening landforms either regionally (Crane and Klimczak, 2019), for a subset of structures (Watters et al., 2015), or globally in large geographic bins (Byrne et al., 2014). Given the thousands of landforms that have been mapped on Mercury, there is an opportunity to study patterns in greater detail. We therefore compile a global tectonic map of fault-related landforms from our previous mapping efforts, extract orientations of all mapped fault segments, and compute their orientations. With this wealth of data, it is possible to study fault orientations with high spatial coverage across Mercury. Although many studies have advanced our understanding of fault orientations on Mercury, orientation data were previously exclusively extracted in latitudinal and/or longitudinal geographic bins, with findings interpreted in the context of latitudinal and/or longitudinal orientations. To maximize the information contained in the compiled data and to search for different patterns, we expand the search for patterns beyond the traditionally used latitudinal and/or longitudinal assessment of fault orientations to also examine geographic bins that follow the geometry of specific tectonic or geological processes.

2. Methodology

2.1. Compilation of the tectonic map

In a geographic information system (GIS), we compiled a dataset containing tectonic structures from the survey of global shortening structures by Byrne et al. (2014), shortening structures in Borealis Planitia from Crane and Klimczak (2019), structures within large impact structures and peak-ring basins (Byrne et al., 2018), and troughs interpreted to be grabens contained within volcanically infilled craters, i.e., ghost craters (Klimczak et al., 2012). We note that other mapping efforts also recorded Mercury's shortening structures (e.g., Man et al., 2023). As for any geological mapping, differences in mapping are expected. Because mapping relies on the same photogeological basemaps and given the number of mapped structures, differences are minor and not expected to lead to different interpretations. Our compiled dataset is publicly available (Klimczak et al., 2025), and contains a total of 18,541

line features representing the traces of surface breaks of faults specified as contractional landforms ($n = 9,845$), extensional landforms ($n = 7,177$), or troughs ($n = 1,429$). Shortening landforms mapped in Borealis Planitia by Byrne et al. (2014) were replaced with those mapped by Crane and Klimczak (2019) to include greater detail in this region. The full dataset is displayed in Fig. 1, showing landforms accommodating shortening and extension in magenta and beige, respectively.

2.2. Data extraction

Because our compiled map contains linework from different mappers, with mapping carried out at multiple scales, we simplified the line features to make them consistent with one another. We applied the “Simplify Line” tool in the ArcGIS toolbox to all polylines in our study to determine and remove vertices that did not contribute to defining the essential shapes of fault surface breaks. For this simplification, we use the algorithm by Douglas and Peucker (1973) with a tolerance of 2000 m, where vertices along the actual mapped line closer to a trend line than the tolerance are eliminated. This process allowed us to identify sections of faults where only two vertices are necessary to sufficiently describe its orientation, resulting in straight fault segments that did not fundamentally differ from the originally mapped polylines. Next, all polylines were split at the remaining vertices to separate the straight fault segments, such that geospatial information could be computed from each of the segments. We calculated geodesic lengths, coordinates of center points, and azimuths for each fault segment. This dataset contains a total of 41,863 fault segments, with 31,479 segments belonging to contractional landforms and 8,845 and 1,538 for extensional landforms and troughs, respectively.

To extract data from specified regions across Mercury, we divided the map area into geographic bins with three different binning systems. Specifically, we divided Mercury into different radial and concentric bins around a center point of geological or geophysical importance. (We discuss the rationale for selecting these points and resulting configurations of geographical binning with the results in Section 3.) First, we divided Mercury into $30^\circ \times 30^\circ$ radial and concentric bins around the present rotational poles, resulting in a total of 72 bins with latitudinal

and longitudinal boundaries. Second, we defined $30^\circ \times 30^\circ$ radial and concentric bins around the center of the Caloris basin at 32°N and 162.5°E , leading to 60 bins arranged around the Caloris basin with basin-radial and -concentric boundaries, as well as two bins for the basin itself and its antipode. Because we do not subdivide the basin and antipodal regions, and because the diameter of the Caloris basin spans more than 30° of arc, the number of bins differs from those in the latitudinal and longitudinal binning. Third, we defined $30^\circ \times 30^\circ$ radial and concentric bins around Mercury’s hot poles (Soter and Ulrichs, 1967) at 0°N , 0°E and 0°N , 180°E , which produces 50 bins, where the two hot poles are not further subdivided and thus lowering the total number of bins. The geographic binning for the Caloris basin and hot pole geometries is publicly available (Klimczak et al., 2025).

For each configuration, radial and concentric bins were created by constructing lines and circles as “Geodesic Features” in the “Advanced Editing” toolbar of ArcGIS. Once all geodesic features were constructed in the desired arrangement, we used the “Features to Polygons” tool to convert the regions bounded by polylines to polygons themselves, and assigned a unique identifier (i.e., a number) to each. We then applied the “Spatial Join” tool to assign the unique identifier of the polygon to the fault segments. Once all fault segments were assigned the geographic bin in which their center points occurred, the data containing fault type, length, coordinates of center point, azimuth, and geographic bin for each fault segment were exported for statistical evaluation and visualization.

2.3. Statistical evaluation of data

When describing orientations of structures anticipated to have developed because of multiple processes or with multiple potential preferred orientations, the best choice of descriptive statistic is the mode, or modes, of the orientation distributions. Identifying modes within orientation distributions allows us to determine the most common structural orientations, unlike computing an average, which could result in misinterpretation of data (as might happen for a bimodal distribution of structure orientations, the average for which would reflect neither of the dominant orientations). To determine statistically

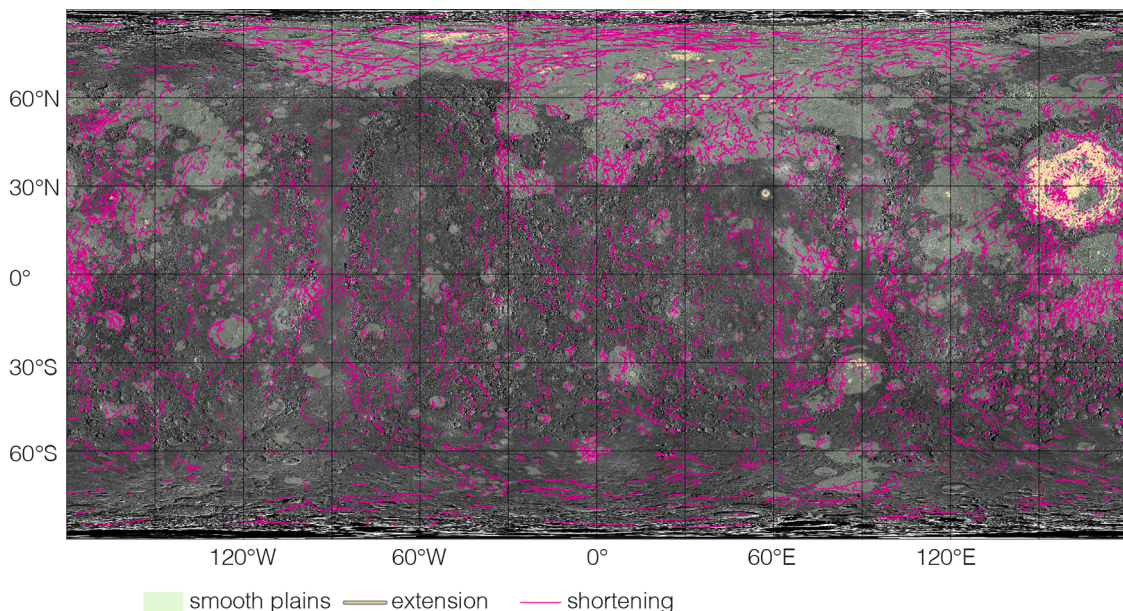


Fig. 1. A global tectonic map of Mercury compiled from global shortening structures by Byrne et al. (2014), shortening structures in Borealis Planitia mapped by Crane and Klimczak (2019), structures within large impact structures and peak-ring basins (Byrne et al., 2018), and troughs within ghost craters (Klimczak et al., 2012). All shortening structures are shown in magenta, all structures accommodating extension are shown in beige, and for reference the expanses of volcanic smooth plains units (Denevi et al., 2013) are shaded in grayish green. The basemap is a monochrome morphology mosaic displayed in equirectangular projection centered at 0° longitude.

significant modes for the binned fault segments, we computed a kernel density estimation function and visualized peaks in the orientation distribution.

Kernel density estimation (KDE) is a non-parametric method of using weighted data to determine the probability of any individual data value (Węglarczyk, 2018, and references therein). Where a histogram displays frequency data binned within the range of values, and a probability density histogram conveys the probability of selecting values within each bin, a KDE produces a probability density function that smooths the latter histogram. This smoothed distribution, produced by summing Gaussian curves centered at each observation (i.e., measured orientation), shows the probability of selecting a value within the range of values. Clustered observations result in greater curve summation and peaks in the KDE. The standard deviation of the individual Gaussian curves, called the bandwidth, is similar in concept to the width of bins in a histogram, and so the choice of bandwidth strongly influences the number and strength of modes detected.

We applied Silverman's Rule of Thumb (Silverman, 1986) in R to calculate the optimal bandwidth from the size, standard deviation, and interquartile range of the entire dataset. Silverman's method tends to over-smooth KDEs. Therefore, we cross-validated our bandwidth choice by also calculating the critical bandwidth with the Improved Sheather and Jones Method (Sheather and Jones, 1991), which typically results in smaller bandwidths. We compared modes calculated with both techniques. Applying the Multimode Package in R (Ameijeiras-Alonso et al., 2021), we were able to use the critical bandwidths to calculate the quantity and values (i.e., orientations) of statistically significant modes for each bin of fault segments. We used the "Locmodes" function (Ameijeiras-Alonso et al., 2019) to determine the respective strength of each mode from the KDE. Both Silverman's Rule and the Sheather and Jones Method produced nearly the same mode values, even if more modes were identified with the latter technique. We tabulated the modes produced by Silverman's Rule (Tables S1 to S3).

We also aimed to visualize the orientation data using rose diagrams; however, binning fault segment orientations in arbitrary degree-width bins ran counter to our goal of searching for important but previously unidentified trends in the data. To determine the degree-width of bins in the rose diagrams, we divided the maximum range of the data (180°) by the critical bandwidth determined by the Silverman's Rule, and rounded to the nearest even integer. This procedure aided in aligning our kernel density distributions with our rose diagrams and allowed each rose diagram to best represent the orientations of the fault segments in that region.

Next, we calculated the cumulative length of all fault segments contained in each of our geographic bins and divided those values by the spherical area that each of our geographic bins occupied. The resulting value provides a measure of structure density for each bin. Finally, we computed length-weighted rose diagrams (Pusztai, 2017) using the previously calculated optimal bin degree-widths for the azimuth data to maximize the visualization of preferred orientations. We color coded each rose diagram according to the structure density found within the geographic bin and plotted them in the center of that bin to create global maps of orientations distributions that maximize our ability to identify map patterns in the distribution of tectonic structures.

3. Results

3.1. Latitudinal and longitudinal patterns

Radial and concentric binning around Mercury's rotational poles allows us to search for a global tectonic pattern along latitudes and longitudes. Studying fault orientations in this arrangement is optimal when considering processes relating to Mercury's spin axis that would produce latitudinal and/or longitudinal patterns, such as those predicted from tidal despinning. This geometric configuration is also the only one that has previously been used for the interpretation of tectonic

patterns on Mercury, for example in broad latitudinal bands (Byrne et al., 2014), as dominant orientations of Mercury's largest structures (Watters et al., 2015), or on a regional scale around the north pole (Crane and Klimczak, 2019). To expand upon the coverage and resolution of these earlier studies, we compute the full set of orientations from all fault-related landforms in our dataset and display them as rose diagrams within six latitudinal bands, 12 longitudinal bands, and all 72 geographic bins that result from dividing Mercury into $30^\circ \times 30^\circ$ bins (Fig. 2). All of our investigated geographical bins return statistically significant modes (Table S1).

Overall orientations observed in entire latitudinal and longitudinal bands are displayed in magenta (Fig. 2). In particular, rose diagrams displayed in magenta on top of Fig. 2 show the entirety of azimuth distributions within longitudinal bands, and those on the right side of Fig. 2 show the entire set of azimuthal observations grouped in latitudinal bands. The map lacks east–west-oriented structures and has a predominance of north–south-oriented structures at the equator and mid-latitudes, shown by the magenta rose diagrams between 60°N and 60°S on the right-hand side of Fig. 2. This observation holds true across all longitudes (Fig. 2; magenta rose diagrams on top). At the poles, all orientations of structures are found but with a slight preference of northwest–southeast directions in the north polar region and northeast–southwest directions in the south polar region, as seen in the magenta rose diagrams north of 60°N and south of 60°S on the right-hand side of Fig. 2, respectively.

The 72 rose diagrams displayed in the center of Fig. 2 show azimuthal distributions of all fault segments located within each bin, with bin boundaries marked by black solid lines and color coding of rose diagrams indicating the spatial density of structures observed in the bins. The lack of east–west-oriented structures and the predominance of north–south-oriented structures at the equator and mid-latitudes is also apparent in this representation. There are differences between the north and south polar regions because of a higher structure density in Borealis Planitia in the north, but generally both poles have many bins with strongly preferred orientations of faults that are not resolved in the rose diagrams combining all observations in those latitudes. The map also shows substantial variation in the orientation data, as evinced by the bi- or even multimodal distributions of many rose diagrams and by noticeable changes in preferred orientations between the individual bins (Figs. 2, S1). Overall, our observations confirm that there is a systematic north–south pattern of fault orientations in the low- and mid-latitudes of Mercury, but that many additional preferred orientations are present that deviate from this pattern.

3.2. Radial and concentric patterns around the Caloris basin

Radial and concentric binning around the Caloris basin is ideal for searching for tectonic patterns related to the impact that formed this basin, as impact damage zones are known to contain fractures of radial and concentric orientations around the impact site (Collins et al., 2004; Xia and Ahrens, 2001). Such fractures may include large-scale joints or normal faults, but we refer to the totality of structure types in the damage zone as *fractures* for simplicity. With a diameter of ~ 1550 km, the Caloris basin is the largest well-preserved basin on Mercury (Murchie et al., 2008) and thus it is likely to retain its extensive damage zone fracture network. Although many of the fractures in such a damage zone are not expected to be large-scale faults, like those analyzed across Mercury with our map data (Fig. 1), fractures introduced by the impact constitute structural discontinuities that have the potential to be reactivated at a later time and be utilized as fault planes from other tectonic processes, such as global contraction. Under such a scenario, thrust faults would grow to produce sizeable shortening landforms with radial and concentric orientations around the Caloris basin. Therefore, investigating fault orientations with this geometric configuration can help us detect the presence and spatial extent of the impact damage zone around the basin. Thus, we compute the full set of orientations from all

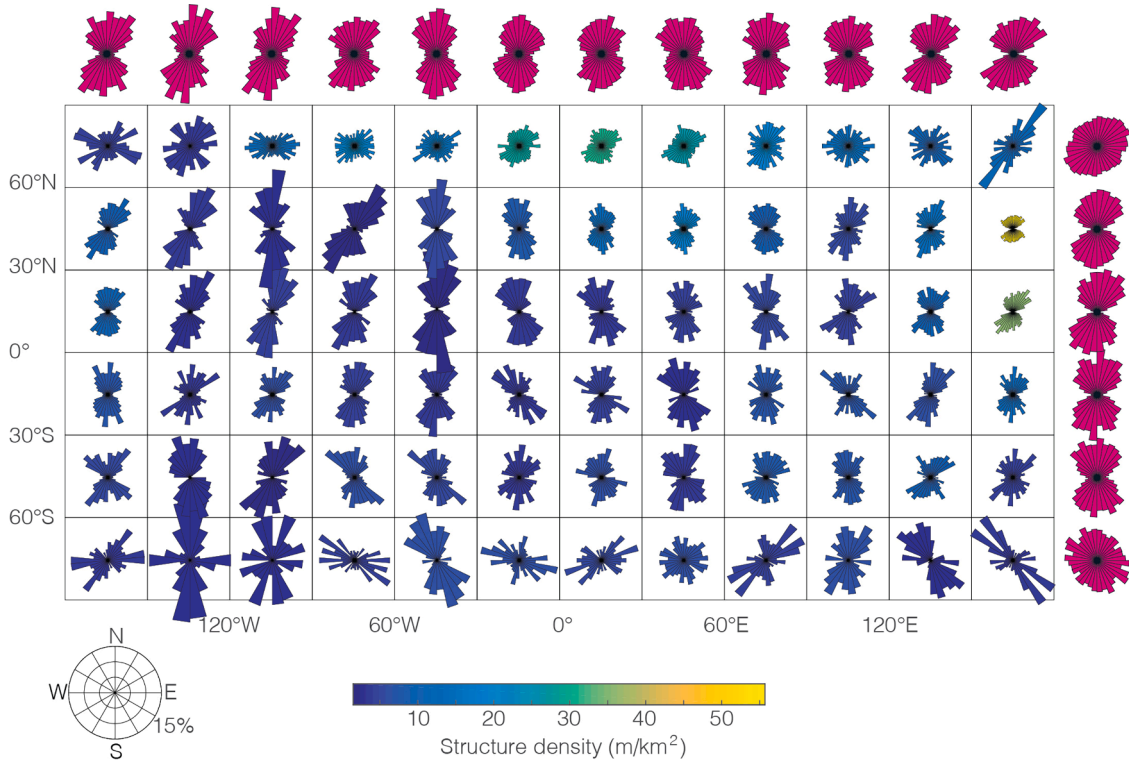


Fig. 2. The distributions of fault orientations shown as rose diagrams in longitudinal and latitudinal arrangements. Magenta rose diagrams along the top display longitudinal variations of fault orientations across all latitudes. Magenta rose diagrams down the right-hand side display latitudinal variations of fault orientations across all longitudes. Rose diagrams showing color coding from blue to yellow indicating structure areal density are displayed in $30^\circ \times 30^\circ$ bins separated by latitudinal and longitudinal boundaries (black lines). All rose diagrams are scaled to one another; the scale is displayed at bottom left. The outer perimeter of the scale explains 15% of all observations. Note that rose diagrams have different bin degree-widths as determined by Silverman's Rule, which results in more bins and smaller diagrams with higher structure densities. The map is displayed in equirectangular projection centered at 0° longitude.

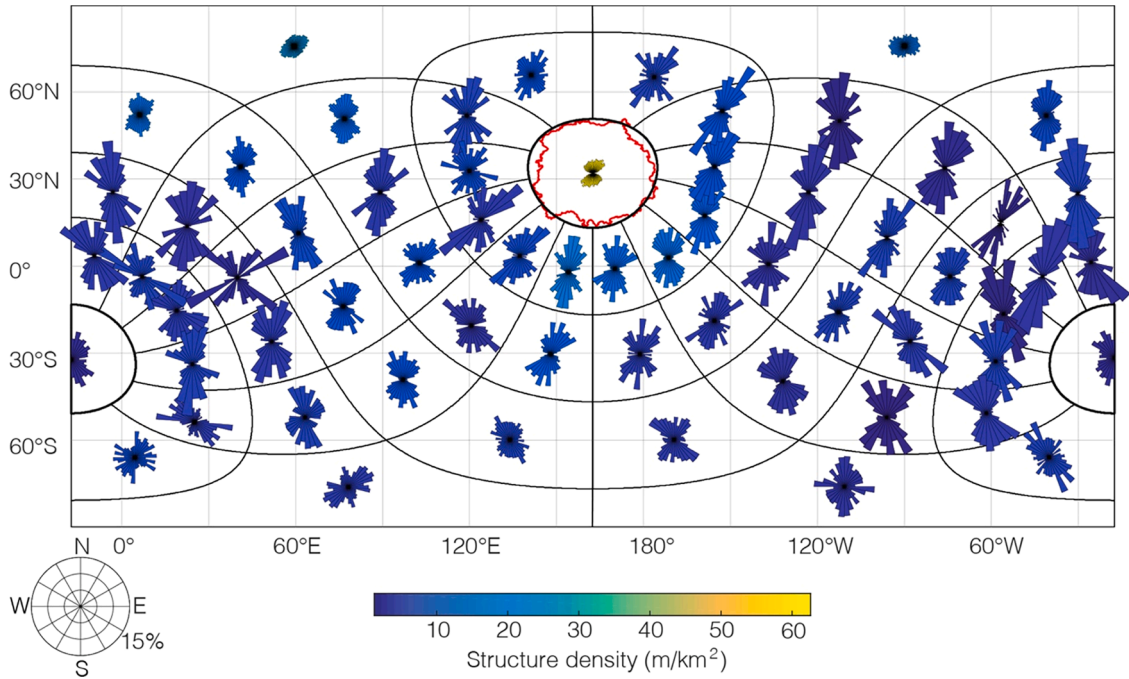


Fig. 3. The distributions of fault orientations shown as rose diagrams in arrangements around the Caloris basin in $30^\circ \times 30^\circ$ bins separated by basin-concentric and -radial boundaries (black lines). Our approximate mapping of the basin outline is plotted in red. The geographic regions outlining the basin and its antipode are bolded. Color coding and scale are the same as in Fig. 2, and rose diagram bin degree-widths again vary across map. The map is displayed in equirectangular projection centered at 162.5° E longitude.

fault-related landforms in our dataset and display them as rose diagrams within 60 bins arranged around Caloris with basin-radial and -concentric boundaries, and two bins for the basin itself and its antipode (Fig. 3). Basin-radial boundaries begin and terminate at the bins outlining the basin and antipodal region (Fig. 3, regions outlined in bold), whereas basin-concentric bin boundaries are those that intersect with the radial boundaries.

The 62 rose diagrams displayed in Fig. 3 are plotted like those in Fig. 2 but reconfigured to show the azimuthal distributions of all fault segments in bins centered around the Caloris basin. As before, all bins contain statistically significant modes (Table S2). Similar to our observations of latitudinal and longitudinal binning (Fig. 2), this Caloris-focused arrangement of map and data also shows few east–west-oriented structures (Fig. 3). Whereas the previously recognized north–south fabric in low- and mid-latitudes is somewhat apparent in this map too, many rose diagrams also show either radial, concentric, or bimodal distributions of radial and concentric preferred fault orientations around the Caloris basin (Figs. 3, S2). Such orientations are most noticeable in the hemisphere in which the basin was emplaced, but the antipodal hemisphere also shows some orientations consistent with basin-radial and -concentric orientations. That prominent orientations radial and/or concentric to the Caloris basin are found to be widespread across Mercury may indicate that the basin-forming impact produced an extensive damage zone in which fractures were subsequently reactivated by faulting.

We also note that, because the Caloris basin itself contains a wealth of fault-related landforms, it has the highest areal density of structures in this representation of our data. The structures within the basin are known to be oriented in basin-radial and -concentric patterns (e.g., Byrne et al., 2018; Cunje and Ghent, 2016), but we consider their formation within the Caloris Planitia as unrelated to fault reactivation of impact damage. The combined data within the basin show a lack of east–west-oriented structures, mirroring global observations. Furthermore, there are preferred orientations in some areas across Mercury, especially noticeable in the southwestern antipodal hemisphere, that

neither fit north–south nor basin-radial and -concentric patterns (Fig. 3).

3.3. Radial and concentric patterns in and around Mercury's hot poles

Mercury has a highly eccentric orbit and is in a spin–orbit resonance with the Sun. Paired with its near-zero obliquity, this orbital configuration leads to the Sun always being overhead at perihelion at the equator at longitudes 0° and 180°. The planet therefore experiences equatorial day–night temperature variations of as much as 600°C at those longitudes, forming the so-called hot poles at these coordinates and warm poles with lower day–night temperature variations in longitudinal bands at 90° and 270° (Soter and Ulrichs, 1967). The temperature variations across the surface (e.g., Vasavada et al., 1999) are found to cause differences in lithospheric strength and thickness (Williams et al., 2011) that have the potential to affect stress states and thus influence fault orientations. Furthermore, these same orbital characteristics also produce solar tides with large tidal displacements (Van Hoolst and Jacobs, 2003). Although global contraction certainly drives faulting, tidal stresses, overlapping with those from global contraction, could influence fault orientation. The detection of a pattern around the hot poles would have important implications for the evolution of the spin–orbit resonance as the known timing and rate of thrust faulting from global contraction (Crane and Klimczak, 2017) then bounds the timing of the acquisition of the modern spin–orbit resonance. Therefore, we investigate the orientations of faults in radial and concentric binning around the hot poles (Fig. 4).

The 50 rose diagrams displayed in Fig. 4 all contain statistically significant modes (Table S3), and a noticeable portion of the azimuthal distributions of fault segments is explained by radial and concentric orientations around the hot poles (Fig. S3). As with Figs. 2 and 3, black solid lines are the bin boundaries and the color coding of rose diagrams indicate the spatial density of structures observed in the bin. This representation of the data once again highlights our previous results showing a lack of east–west-oriented structures and a predominance of north–south-oriented structures at the equator and mid-latitudes, as well

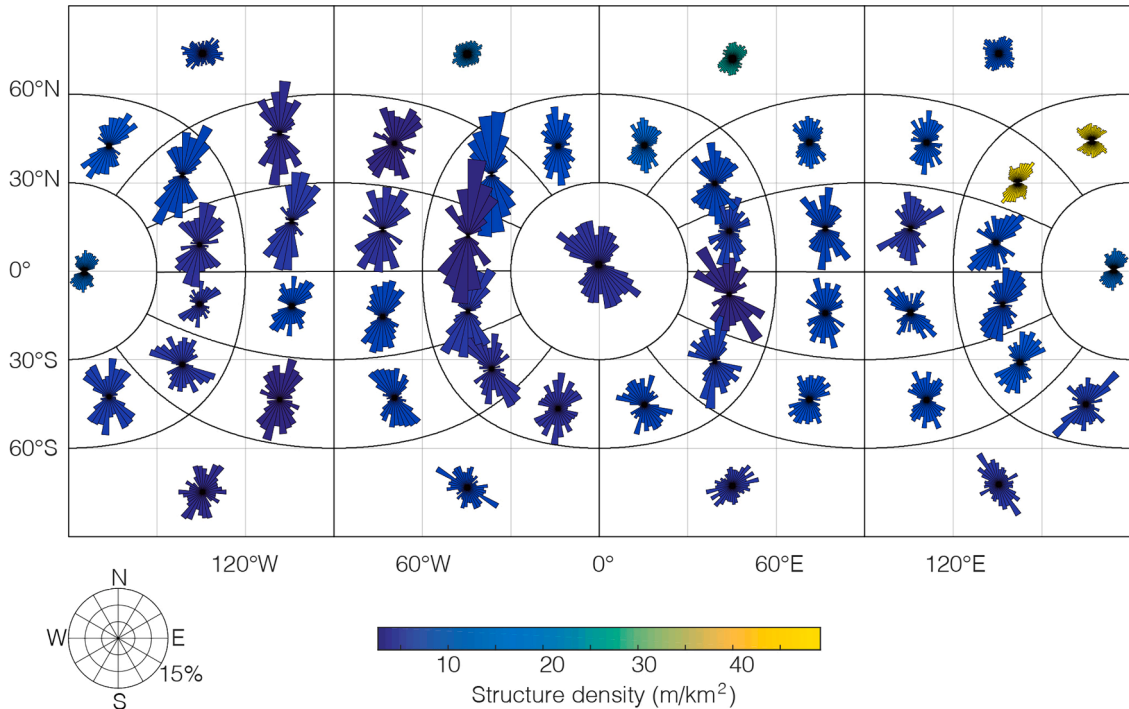


Fig. 4. The distributions of fault orientations shown as rose diagrams in 30° × 30° bins separated by boundaries radial and concentric around the hot poles (black lines). Color-coding and scale are the same as in Fig. 2, and rose diagram bin degree-widths again vary across map. The map is displayed in equirectangular projection centered at 0° longitude.

as the greater number of structures in the Caloris basin. Faults at the hot poles show all ranges of orientations from northwest–southeast to northeast–southwest, and follow the general global trend of a relative dearth of east–west-oriented structures. Interestingly, there are systematic variations in preferred orientations of faults found concentrically around the hot poles, a pattern that is best noticed in the southern hemisphere in the geographic bins that connect the warm poles with the cold (rotational) poles, outside the densely faulted Caloris and Borealis Planitiae. This finding establishes that there is a pattern of tectonic shortening strain around the hot poles.

4. Discussion

Analysis of the orientations of nearly 42,000 fault segments across Mercury obtained from our compilation of a global tectonic map (Fig. 1) shows clear evidence of multiple, superposed tectonic patterns on the planet, which generally confirms the lineament orientations analyzed on Mariner 10 images (e.g., Thomas et al., 1988). Most obviously, our data show that a north–south pattern of shortening landforms exists in low- and mid-latitudes. This pattern is obvious in all our spatial representations of the data (Figs. 2–4). However, we also find bi- and multimodal distributions of orientations in our rose diagrams, indicating that more than one preferred orientation of structures is represented within many of the investigated geographic regions. In fact, some geographic regions show up to five statistical modes (Tables S1–S3), indicating that a single process may have produced more than one preferred orientation, that as many as perhaps five processes overlapped to form the observed multi-modal distributions, or that the observations are explained by a combination of these possibilities. Systematic changes in preferred orientations across Mercury become apparent when the data are analyzed in different geographic configurations. In particular, preferred orientations are found to follow radial and concentric geometries around the Caloris basin (Figs. 3, S2) and the hot poles (Figs. 4, S3), showing that certain tectonic patterns may only become apparent when fault orientations are studied in spatial arrangements tied to geological or geophysical phenomena.

Because multiple patterns are superposed, some geographic bins will show dominant orientations of one pattern even when the data are analyzed in a spatial arrangement chosen to highlight another pattern. It is equally possible that a preferred orientation in a geographic region is consistent with multiple processes. Thus, in regions where there is a strong overlap of two or more patterns, it is important to not focus solely on the strongest mode when interpreting the data. In fact, some modes are produced by our statistical evaluation with low mode strengths, but they represent local maxima in the data. Such modes are not always prominently represented within the rose diagrams, especially when the data are dominated by another mode. Therefore, visual assessment of the map and statistical techniques must be applied together to interpret any superposed patterns.

We identified multiple, superposed global tectonic patterns by specifically investigating three different configurations of the data and included all types of structures that were part of our tectonic map. The global patterns we studied are unlikely to be affected by or result from the combining of all structure types, as the vast majority of structures are shortening landforms, with those landforms accommodating extension being much smaller and only occurring in volcanic units within some impact structures (Fig. 1). Regional patterns are likely to emerge when data are further analyzed by fault type and with a different geographic binning, such as distinguishing orientation data within different geological units. The multi-modal distributions found in our analysis indicate that additional unidentified global or regional tectonic patterns with different geographic configurations are also likely to be present on Mercury. Thus, our method will be useful for future studies of tectonic patterns and their underlying causes on Mercury—and may further be applied to any planetary body on which a multitude of tectonic landforms is found.

Mercury's north–south pattern of shortening landforms in the equatorial and mid-latitudinal regions (Fig. 2) is consistent with the preferential illumination of linear landforms from east and west solar azimuth directions inherent to all MESSENGER photogeology (Byrne et al., 2018). Although Watters et al. (2015) argued that such lighting bias is unlikely to affect the recognition of the largest structures, Fegan et al. (2017) found that their mapping was influenced by illumination geometry. Until topographic datasets capable of resolving tectonic structures (e.g., Preusker et al., 2017) become available globally to allow mapping efforts based on photogeology to be complemented with artificially illuminated hillshade images, it remains an open question as to how far lighting bias influences the interpretation of the north–south orientations in Mercury's low- and mid-latitudes. This pattern is, however, equally consistent with predictions of tidal despinning overlapping with global contraction (Beuthe, 2010; Klimczak et al., 2015; Pechmann and Melosh, 1979). Because detailed discussions on the pattern, the timing, and likelihood of overlap of these two processes are offered by Klimczak et al. (2015) and Byrne et al. (2018), we focus our discussion here on those patterns that we detected to be superposed on top of the dominant north–south trends. Any patterns that do not show that strong north–south orientation—especially if they show latitudinal variations of orientations—are unlikely to reflect biases arising from the particular illumination conditions at Mercury.

Shortening landforms and their underlying thrust faults show preferential orientations radial and concentric to the Caloris basin, especially in the hemisphere in which this large basin was emplaced (Fig. 3). The distribution and orientations of such structures both are consistent with the geometry of an extensive impact damage zone. Under such a scenario, the Caloris impact would have produced shock waves that propagated radially away from the site of impact, causing rock fracturing in radial and concentric geometries around the basin. Field observations and numerical modeling both show that, for medium-sized impacts, the extent of impact damage zones near the surface is a factor of ~2–3 greater than the radius of the final crater (Gurov et al., 2007; Collins et al., 2004, 2008). Whereas the formation of the basin itself has been modeled (Gosselin et al., 2023; Potter and Head, 2017), damage zone sizes for large basins like Caloris are not well understood. Yet it is not unreasonable for the damage of large basins to be widespread, affecting an entire hemisphere. The size of this damage zone and the dominant role it played in the tectonic history of Mercury is comparable to the Tharsis rise on Mars, making it a tectonic province that affected the structure of the innermost planet far beyond the extent of the Caloris basin itself.

The faults and joints within such an impact damage zone likely constituted prominent structural discontinuities in Mercury's lithosphere, some of which were suitably oriented to be reactivated by global contraction to become thrust faults and growing into the large shortening landforms we observe. Frictional sliding theory requires pre-existing planes of weakness such as fractures to be present in a lithosphere to be (re-)activated as faults (Byerlee, 1967). That such a widespread tectonic pattern consistent with impact damage from the Caloris basin impact is observed on Mercury attests to the important role that fractures in impact damage zones play for the formation of large-scale landforms in planetary tectonic settings. Many other fault-related landforms on Mercury then may have originated as a structural discontinuities from large impact events, which may explain some of the variation of orientations in our fault dataset. Future investigations of structural orientations around the population of other large impact basins (Fassett et al., 2012) may reveal additional patterns.

We detect systematic variations of preferred orientations of faults concentrically around the hot poles, a pattern that is most noticeable in the anti-Caloris hemisphere, especially in the geographic bins coincident with the warm and cold poles (Figs. 4, S3). Preferred orientations around the hot poles could be caused either by fault growth influenced by insulation-driven lithospheric strength and thickness variations, or by stresses arising from solar tides overlapping with those from global

contraction to produce a difference in the horizontal principal stress components that, in turn, would have led to preferred instead of random fault orientations.

The outputs of lithospheric evolution models are strongly affected by the temperature distributions across Mercury (Williams et al., 2011; Fleury et al., 2024), whereby the lithosphere is stronger and thicker at the cold and warm poles than the hot poles (Williams et al., 2011). These results were interpreted to cause the preferential orientation and distribution of shortening landforms (Williams et al., 2011). Our results indicate that mapped structure orientations are indeed consistent with the pattern of lateral strength variations across Mercury, but that differences in structure areal density—which could be indicative of preferential distribution following this same pattern—are too subtle in our representation of the data (Fig. 4) to reveal any unequivocal correlation.

Mercury's hot poles also coincide with its tidal axis (Matsuyama and Nimmo, 2009), and so it is possible that tectonic patterns arise from tidal stresses symmetric around this axis. Under this scenario, global contraction would drive the faulting, as tidal stresses alone are likely not sufficient to initiate or sustain faulting on Mercury. Global contraction, when acting alone, produces horizontally isotropic compressive stresses (Melosh and McKinnon, 1988) that cause random orientations of thrust faults. An overlap with stresses raised by solar tides would change the magnitudes of horizontal stress components, which could lead to preferential thrust fault orientations symmetric about the tidal axis.

In addition, tidal displacements are repeated in numerous cycles over geological time, such that fatiguing of the lithosphere may serve to reduce its strength that, in turn, could also be reflected in a tectonic pattern. Fatiguing facilitates the initiation and propagation of micro-scale cracks in a material due to cyclic loading. Once a fatigue crack has formed, sufficient subsequent stress propagates the crack a small amount with each loading cycle. Mercury's tides are calculated to cause surface displacements with amplitudes of up to 2 m at the tidal axis and a period of about 88 Earth days, corresponding to one Mercury year (Van Hoolst and Jacobs, 2003). Repeated over millions of years, Mercury's tidal cycle may have produced preferentially oriented microscale cracks that then grew into larger fractures with preferential orientations that were utilized as thrust faults from global contraction. In addition, fatiguing may have weakened the lithosphere in areas affected most by the microscale cracking, causing systematic strength variations throughout Mercury's lithosphere.

Whether the observed fault orientations around Mercury's hot poles were formed by variations in insulation- or fatiguing-influenced variations in lithospheric strength, by tidal stresses overlapping with global contraction, or perhaps by a combination of any or all of these factors, remains to be investigated in detail. But a tectonic pattern resulting from any of these scenarios has implications for Mercury's 3:2 spin-orbital resonance with the Sun. Many of the landforms that form this pattern are billions of years old, with thrust faulting being well underway by the Calorian period (Crane and Klimczak, 2017) sometime after 3.9 to 3.5 Ga ago (Denevi et al., 2018), thus requiring that resonance to have been established at or before the onset of faulting. The origin and evolution of the 3:2 spin-orbit resonance has been discussed (e.g., Correia and Laskar, 2009, 2004; Noyelles et al., 2014; Wiczeorek et al., 2012), and modeling that incorporates geophysical constraints from MESSENGER observations suggests that Mercury acquired this resonance very early in its history (Noyelles et al., 2014), consistent with our observations. Future orbital evolution models of Mercury may thus use the tectonic pattern around the hot poles to place a temporal constraint on the acquisition of the present spin-orbit resonance and, indeed, to test whether any earlier resonances, should they have existed, might be reflected in the planet's tectonic inventory.

5. Conclusions

We extracted orientation data from a compiled global tectonic map of Mercury, a dataset containing 18,541 fault-related landforms (Fig. 1)

that is publicly available (Klimczak et al., 2025). We observe a large variability in the data across Mercury where many fault orientations are not explained by the previously established north-south-oriented thrust faults in the low- and mid-latitudes. We find that fault orientations display bi- or multi-modal distributions, in places showing up to five modes (Tables S1–S3), indicating that several tectonic processes governed the orientations of faults on Mercury. Our results demonstrate that tectonic patterns are effectively visualized and interpreted when analyzed with a geographic binning that follows the geometry of a known or hypothesized geological phenomenon. We specifically investigated radial and concentric configurations around the rotational poles (Figs. 2, S1), the Caloris basin (Figs. 3, S2), and the hot poles (Figs. 4, S3), all of which show preferred orientations that match the particular geometry of the geographic binning. More patterns may be detected in Mercury's fault dataset in future studies that carry out in-depth analyses of fault orientations by, for example, lithologic units and fault type. Although we applied our study to Mercury's fault-related landforms, this methodology is applicable to any structural mapping on any planetary body hosting tectonic landforms, and may thus be used to gain better insight into tectonic processes across the Solar System more generally.

CRediT authorship contribution statement

Christian Klimczak: Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kelsey T. Crane:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **Paul K. Byrne:** Writing – review & editing, Validation, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.epsl.2025.119331](https://doi.org/10.1016/j.epsl.2025.119331).

Data availability

The shapefile of the compiled tectonic map is available as open-access dataset: Klimczak, C., Byrne, P. K., and Crane, K. T. (2025): A Global Tectonic Map of Mercury. Mendeley Data, V2, [doi:10.17632/p43b9wttjp.2](https://doi.org/10.17632/p43b9wttjp.2) (<https://data.mendeley.com/datasets/p43b9wttjp/2>). The shapefile of our interpreted outline of the Caloris basin is available as part of the open-access dataset in Byrne, P. K., and Klimczak, C. (2025): A Tectonic Map of the Caloris Basin, Mercury. Mendeley Data, V1, [doi:10.17632/jxzhdvf5xr.1](https://doi.org/10.17632/jxzhdvf5xr.1) (<https://data.mendeley.com/datasets/jxzhdvf5xr.1>). We plot our length-weighted rose diagrams using a MATLAB script: Pusztai, A. (2017): Weighted rose. MATLAB Central File Exchange (<https://www.mathworks.com/matlabcentral/fileexchange/65009-weighted-rose>).

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